# 1 I

## 1.1 (1)

The equation of motion in the $\theta$ direction is

$$ml\ddot{\theta} = -mg\sin\theta.$$

Under the small-angle approximation,

$$\ddot{\theta} + \frac{g}{l}\theta = 0.$$

The general solution is

$$\theta(t) = A\cos\left(\sqrt{\frac{g}{l}}t\right) + B\sin\left(\sqrt{\frac{g}{l}}t\right).$$

Since $\theta(0) = \alpha$ is the maximum angular displacement, we also have $\dot{\theta}(0) = 0$. Applying these initial conditions gives

$$A = \alpha, \qquad B = 0.$$

Therefore,

$$\boxed{\theta(t) = \alpha\cos(\omega t), \qquad \omega = \sqrt{\frac{g}{l}}}.$$

## 1.2 (2)

By conservation of mechanical energy, the maximum kinetic energy equals the gravitational potential energy gained at the turning point:

$$E = mgl(1 - \cos\alpha).$$

Using the small-angle approximation,

$$\boxed{E \approx \frac{1}{2}mgl\alpha^2}.$$

## 1.3 (3)

From the equation of motion in the radial direction,

$$T - mg\cos\theta = ml\dot{\theta}^2.$$

Hence,

$$T = mg\cos\theta + ml\dot{\theta}^2.$$

Using the result from (1),

$$\dot{\theta} = -\alpha\omega\sin(\omega t),$$

we obtain

$$ml\dot{\theta}^2 = mg\alpha^2\sin^2(\omega t).$$

Meanwhile, to second order in $\alpha$,

$$\begin{aligned} mg\cos\theta &\approx mg\left(1-\frac{\theta^2}{2}\right) \\ &= mg\left(1-\frac{\alpha^2}{2}\cos^2(\omega t)\right). \end{aligned}$$

Therefore,

$$\boxed{T(t) \approx mg\left[1+\alpha^2\sin^2(\omega t)-\frac{\alpha^2}{2}\cos^2(\omega t)\right]}.$$

### 1.4 (4)

The time average of the tension over one period is

$$\langle T\rangle = \frac{1}{\tau}\int_0^\tau T(t)\,dt, \qquad \tau = \frac{2\pi}{\omega}.$$

Since

$$\langle\sin^2(\omega t)\rangle = \langle\cos^2(\omega t)\rangle = \frac{1}{2},$$

we obtain

$$\begin{aligned} \langle T\rangle &= mg\left(1+\frac{\alpha^2}{2}-\frac{\alpha^2}{4}\right) \\ &= \boxed{mg\left(1+\frac{\alpha^2}{4}\right)}. \end{aligned}$$

## 2 II

### 2.1 (5)

When the string length changes by $\delta l$, the angular frequency becomes

$$\begin{aligned} \omega+\delta\omega &= \sqrt{\frac{g}{l+\delta l}} \\ &= \omega\left(1+\frac{\delta l}{l}\right)^{-1/2} \\ &\approx \omega\left(1-\frac{\delta l}{2l}\right). \end{aligned}$$

Therefore,

$$\boxed{\delta\omega = -\frac{\omega}{2l}\delta l}.$$

### 2.2 (6)

Since the tension acts toward the pivot, the work done by the tension during a slow change in the string length is

$$\delta W = -\langle T\rangle\delta l.$$

Using the result from (4),

$$\boxed{\delta W = -mg\left(1+\frac{\alpha^2}{4}\right)\delta l}.$$

The total mechanical energy consists of the pendulum energy $E$ and the gravitational potential energy at its lowest position.

Taking the gravitational potential energy to be zero at the pivot, the total mechanical energy is

$$E_{\text{tot}} = E - mgl.$$

Thus, by the work-energy relation,

$$\delta W = \delta E - mg\delta l,$$

which gives

$$\begin{aligned}\delta E &= mg\delta l + \delta W \\ &= mg\delta l - mg\left(1+\frac{\alpha^2}{4}\right)\delta l.\end{aligned}$$

Therefore,

$$\boxed{\delta E = -\frac{1}{4}mg\alpha^2\delta l}.$$

### 2.3 (7)

Combining the results from (2) and (6),

$$E = \frac{1}{2}mgl\alpha^2, \qquad \delta E = -\frac{1}{4}mg\alpha^2\delta l,$$

we obtain

$$\frac{\delta E}{E} = -\frac{\delta l}{2l}.$$

Meanwhile, from (5),

$$\frac{\delta\omega}{\omega} = -\frac{\delta l}{2l}.$$

Thus,

$$\frac{\delta E}{E} = \frac{\delta\omega}{\omega}.$$

It follows that

$$\begin{aligned}\delta\left(\frac{E}{\omega}\right) &= \frac{\delta E}{\omega} - \frac{E}{\omega^2}\delta\omega \\ &= \frac{E}{\omega}\left(\frac{\delta E}{E} - \frac{\delta\omega}{\omega}\right) \\ &= 0.\end{aligned}$$

Therefore,

$$\boxed{\frac{E}{\omega} = \text{constant}}.$$

# 3 III

## 3.1 (8)

The Lagrangian of the pendulum is

$$\begin{aligned} L &= K - V \\ &= \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1-\cos\theta). \end{aligned}$$

Under the small-angle approximation,

$$L \approx \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}mgl\theta^2.$$

The generalized momentum conjugate to $\theta$ is

$$\boxed{p = \frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}}.$$

## 3.2 (9)

The energy of the pendulum is

$$\begin{aligned} E &= \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1-\cos\theta) \\ &\approx \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}mgl\theta^2 \\ &= \frac{p^2}{2ml^2} + \frac{mgl\theta^2}{2}. \end{aligned}$$

Rearranging the equation gives

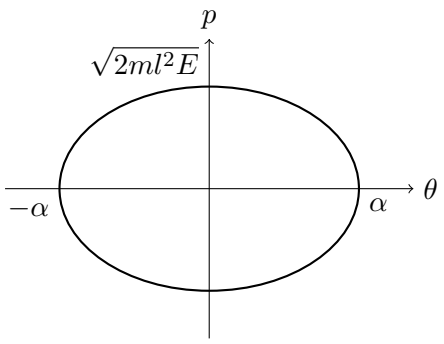
$$\boxed{\frac{p^2}{2ml^2E} + \frac{\theta^2}{2E/(mgl)} = 1}.$$

This is the equation of an ellipse centered at the origin in the $(\theta, p)$ phase space.

The semi-major and semi-minor axes, along the corresponding coordinate directions, are

$$a = \sqrt{\frac{2E}{mgl}} = \alpha, \qquad b = \sqrt{2ml^2E}.$$

Thus, the phase-space trajectory is a closed ellipse with horizontal semi-axis $a$ and vertical semi-axis $b$.



### 3.3 (10)

The integral over one complete oscillation corresponds to the area enclosed by the phase-space trajectory.

Since the trajectory is an ellipse,

$$\oint p\, d\theta = \pi ab.$$

Using the semi-axes obtained in (9),

$$\begin{aligned}\oint p\, d\theta &= \pi\sqrt{\frac{2E}{mgl}}\sqrt{2ml^2E} \\ &= 2\pi E\sqrt{\frac{l}{g}} \\ &= 2\pi\frac{E}{\omega}.\end{aligned}$$

Therefore,

$$\boxed{\oint p\, d\theta = 2\pi\frac{E}{\omega}}.$$

From the result of (7),

$$\delta\left(\frac{E}{\omega}\right) = 0.$$

Hence,

$$\boxed{\delta\left(\oint p\, d\theta\right) = 0}.$$

Thus, the phase-space integral remains invariant under a slow change in the string length. This quantity is known as an adiabatic invariant.